Application Reports 2025

How customers use Alvery pin modules. Customer names are withheld at their request.

Gearbox housings on coordinate measuring machines

Situation

A supplier of aluminium gearbox housings measures 14 housing variants on two coordinate measuring machines.

Before

Each variant had its own fixture. A change-over took up to 40 minutes, and fixtures filled two shelving units.

Solution

One FormRest XL Set (42.3168) per machine. Housings are placed on four pin supports and locked by hand.

Results

- Change-over time reduced from 40 minutes to under 3 minutes
- Repeatability within 0,02 mm over 30 placements
- Payback within 5 months

Turbine blade roots in 5-axis machining

Situation

A manufacturer of turbine components machines blade roots of 9 blade types in small batches.

Before

Soft jaws were milled for each blade type; every new type required two days of jaw design and machining.

Solution

FormVise 140 H modules (41.0674) as vise jaws. The 42 mm pin stroke covers the twisted blade profile.

Results

- No more soft jaws for new blade types
- Set-up time per batch reduced by 70 %
- Clamping torque 25 Nm, holding force 7,5 kN per module, sufficient for roughing

Cobot handling of medical parts

Situation

A medical device maker assembles 23 variants of small plastic housings with a collaborative robot.

Before

The robot needed a gripper change for almost every variant, which limited batch sizes.

Solution

Two AutoForm Mini modules (44.0580) on a parallel gripper. The pins adapt to each housing.

Results

- One gripper for all 23 variants

- Gripper changes eliminated
- Workpiece weights 80–400 g, well below the 1 kg limit

AGV-based final assembly

Situation

An automotive supplier moves seat frames between 6 assembly stations on driverless transport systems.

Before

Each AGV carried variant-specific supports, and the fleet needed 3 support types per vehicle.

Solution

One FormBed e (45.0831) per AGV, locked electrically when the frame is placed.

Results

- One universal support per AGV
- No compressed air needed on the vehicle
- Investment recovered within 11 months